



Sheet 8

Matter Waves

De Broglie Wavelength

1. Calculate the de Broglie wavelength for a proton moving with a speed of 10^6 m/s.
2. Through what potential difference would an electron have to be accelerated to give it a de Broglie wavelength of 1.00×10^{-10} m?
3. An electron and a photon each have kinetic energy equal to 50 keV. What are their de Broglie wavelengths?
4. To “observe” small objects, one measures the diffraction of particles whose de Broglie wavelength is approximately equal to the object’s size. Find the kinetic energy (in electron volts) required for electrons to resolve a nucleus of size 10 fm.
5. Find the de Broglie wavelength of a ball of mass 0.20 kg just before it strikes the Earth after being dropped from a building 50 m tall.

The Heisenberg Uncertainty Principle

6. A proton has a kinetic energy of 1.0 MeV. If its momentum is measured with an uncertainty of 5.0%, what is the minimum uncertainty in its position?
7. Estimate the kinetic energy of an electron confined within a nucleus of size 1.0×10^{-14} m by using the uncertainty principle.
8. A charged pi meson has a rest energy of 140 MeV and a lifetime of 26 ns. Find the minimum energy uncertainty of the pi meson.